

Stage 2

AI System Design PROJECT
Umm Al Qura University



Project Title: Smart Parking detection with proximity recommendations

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Stage 2

AI System Design PROJECT Umm Al Qura University



2.1 Data Strategy and Architecture Design

2.1.1 Identification and Justification of Data Sources

The dataset selected for this project is a pre-processed version of the PKLot dataset available on Kaggle. It is derived from the original PKLot research dataset and adapted to support object detection tasks.

The dataset includes 12,416 images of outdoor parking lots captured by surveillance cameras in three separate locations in Brazil: PUCPR, UFPR04, and UFPR05. The images capture various weather conditions, including sunny, cloudy, and rainy days, providing a robust environment for evaluating computer vision models under different lighting and shadow effects. This variation in lighting and weather conditions helps prevent the model from overfitting to a specific pattern, enabling it to accurately recognize parking spaces across various times and real-world environmental conditions.

Each parking space within an image is manually labeled as "Occupied" or "Empty." The images are in JPEG format, with a resolution of 1280×720. The original annotations have been converted into axis-aligned bounding boxes, which are available in CSV and TXT formats, ensuring that the dataset is fully compatible with standard object detection frameworks.

2.1.2 Data Collection and Cleaning Plan

Data Preprocessing

The preprocessing stage is responsible for preparing and organizing the parking images before they are passed to the YOLOv8 model for training and inference. Proper preprocessing is essential to ensure dataset consistency and support efficient model learning.

The dataset utilized in this project is primarily derived from the PKLot parking dataset. To reduce the time and effort required for manual annotation, a pre-labeled version of the same dataset was obtained from Roboflow Universe. This version includes ready-made annotations for parking-related objects and vehicles using bounding boxes in YOLO format.

The preprocessing pipeline includes the following steps:

- **Dataset Organization:**
The dataset is organized into training, validation, and testing subsets to support model training and evaluation.
- **Data Cleaning:**
The dataset and labels were manually checked to ensure consistency and correct formatting before training.
- **Image Resizing:**
All images are automatically resized to 640×640 pixels to ensure compatibility with the YOLOv8 architecture and maintain consistent input dimensions across the dataset.
- **Label Verification and Preparation:**
Bounding box annotations provided in YOLO format were verified to ensure correct object localization and annotation consistency.

Stage 2

AI System Design PROJECT Umm Al Qura University



The overall objective of preprocessing is to provide clean, standardized, and properly annotated image data that enables accurate and efficient parking occupancy detection using YOLOv8.

2.1.3 Initial System Architecture Diagram

Our system is based on intelligent detection of available parking spaces combined with recommending the nearest parking spot. The system is designed as a sequential series of stages to ensure smooth data flow and allow future development and scalability.

The system begins with the parking dataset stage, which contains images or video clips of parking lots captured under different conditions such as variations in lighting and camera angles. After that, the data is loaded into the working environment for processing.

Next, data cleaning is performed. In this stage, corrupted, distorted, duplicated, or low-quality images are removed to prevent negatively affecting the model's performance and to maintain dataset consistency before the training process.

After that, image preprocessing is applied. All images are resized to a unified fixed size to ensure consistency across the dataset. The dataset is then split into training, validation, and testing sets. During model training and inference, the YOLO model performs internal processing such as normalization automatically; therefore, no additional manual normalization is applied in the preprocessing stage.

Then, the parking space detection stage is performed. In this stage, parking spaces are identified within the image or video using a YOLO object detection model to detect parking spaces and their occupancy status.

After extracting each parking slot individually, the system proceeds to the occupancy classification stage. In this stage, each parking space is analyzed to determine whether it is available or occupied based on the visual information extracted from the image.

Next, the system moves to the decision-making and routing stage. First, only the available parking spaces are filtered from the detected slots. The user's location is mapped directly onto the parking lot image as a visual reference point. To determine the optimal path within the parking layout, the system applies the A* search algorithm, which is used to compute the shortest and most efficient route from the user's position to the nearest available parking space based on a graph representation of the parking structure. The closest vacant slot is then highlighted visually on the interface as the optimal choice.

After selecting the parking space, the system determines the route within the same visual environment, guiding the user through the parking layout using image-based navigation enhanced by the computed A* path rather than external maps or numerical calculations.

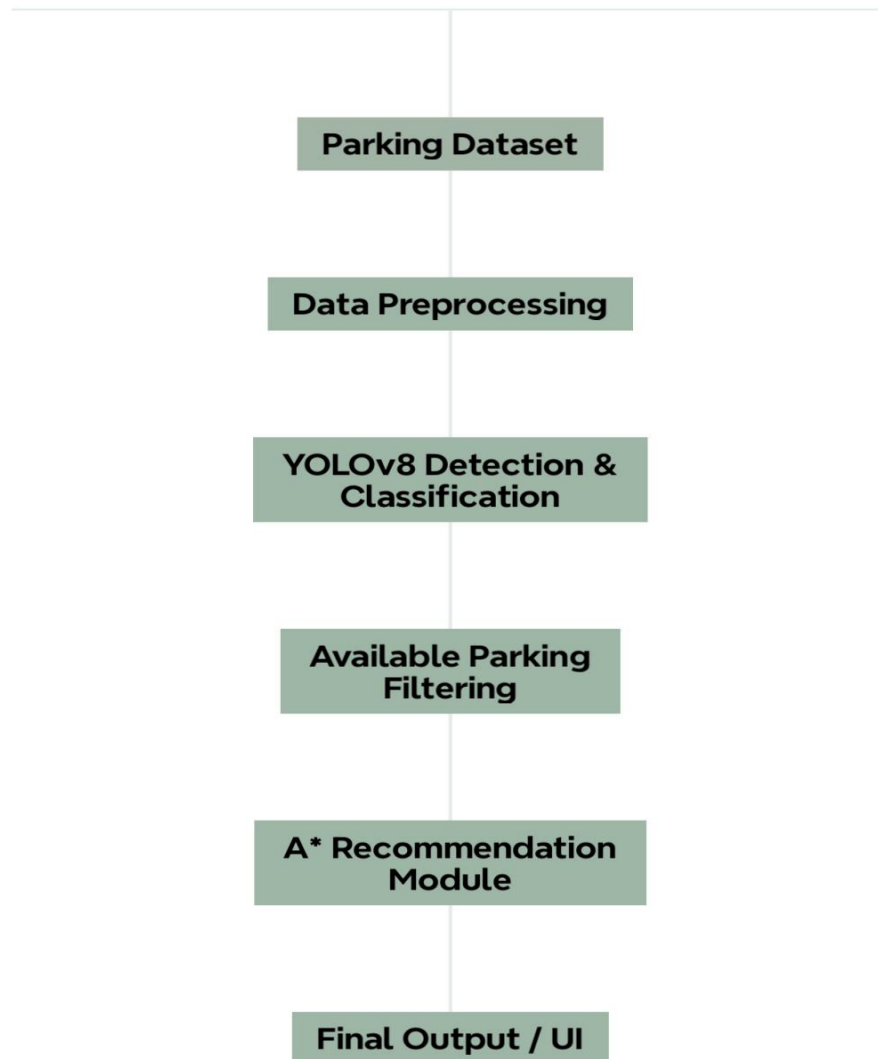
Finally, the results are displayed through the user interface. The parking lot image is shown with occupied and available spaces clearly distinguished, while both the user's position and the nearest available parking space are visually highlighted to provide clear guidance.

Overall, the architectural design allows future scalability, such as integrating live cameras, which makes the system suitable for real-world applications.

Stage 2

AI System Design PROJECT
Umm Al Qura University

diagram System Architecture



Stage 2

AI System Design PROJECT Umm Al Qura University



2.2 Detailed System Design

2.2.1 Logical System Architecture and Components

The proposed system is based on a modular architecture composed of several interconnected components that work together efficiently to produce the final output.

1. Data Pipeline Component:

This is responsible for handling the dataset, such as loading the data, cleaning it, preprocessing it, and splitting it. It ensures that the data is properly prepared for model training and evaluation.

2. Model Component:

This is divided into two stages: Object Detection and Classification. The system uses a deep learning model to detect parking spaces and classify each space as either occupied or available based on visual features extracted from images.

3. Application Layer (User Interface):

Represents the interaction between the user and the system. The results are displayed through a visual interface that shows available parking spaces, highlights the user's position, and marks the nearest available parking spot directly within the parking layout image. The system provides intuitive visual guidance based on the spatial arrangement of the scene.

4. Backend/API Layer:

Handles communication between the user interface and the system components. It receives user inputs such as location data and returns the processed results from the system.

5. Monitoring Component:

This is responsible for monitoring the overall performance and behavior of the system during operation. It tracks model accuracy, system response time, and detects any errors or failures that may occur. It also logs system activities to help in debugging and improving performance over time. This ensures that the system remains reliable, efficient, and performs as expected in real-world usage.

6. Storage Component:

This is responsible for storing the dataset, trained models, and system outputs. It ensures data availability and supports system scalability and future updates.

Stage 2

AI System Design PROJECT Umm Al Qura University



2.2.2 Scalability, Reliability, Fault-Tolerance, and Security Considerations

Scalability

The proposed system can be characterized by high scalability, as it can be further developed in the future to operate on a larger scale without requiring changes to its core architecture. Additional parking areas can be integrated, or the system can be connected to live cameras to obtain real-time data. Moreover, the modular design allows for easy addition or improvement of models and algorithms, making the system suitable for deployment in larger environments such as smart cities while maintaining efficient performance.

Reliability

The system is designed using sequential pipeline architecture, which logically separates the detection process from the pathfinding logic. This structure ensures that the outputs from the visual detection model are passed as organized inputs to the navigation module, making the system easier to maintain and test at each stage.

To ensure stability and consistent performance, the system utilizes the YOLO framework, which is known for its ability to handle various environmental changes. By training the model in a wide range of lighting conditions and parking scenarios, the system achieves the necessary generalization to provide reliable recommendations for the nearest available parking space.

Fault-Tolerance

To improve system reliability and maintain stable performance, several fault tolerance considerations are incorporated into the proposed system. The system is designed to handle issues such as unreadable images, missing labels, or invalid inputs by skipping corrupted data during preprocessing to prevent interruptions during training and inference.

In addition, using a pre-labeled and organized dataset helps reduce annotation inconsistencies and minimizes detection errors. The dataset is also divided into training, validation, and testing subsets to monitor model behavior and reduce the risk of overfitting or unreliable predictions.

Furthermore, YOLOv8 provides confidence scores for detections, allowing low-confidence predictions to be filtered in order to improve output reliability and reduce false detections during real-time operation.

Stage 2

AI System Design PROJECT Umm Al Qura University



Security

A significant advantage of selecting the PKLot dataset is its inherent contribution to system security and privacy; since the images lack sensitive personal identifiers—such as discernible faces or license plates—the risk of data leakage or privacy breaches is mitigated at the source [4]. This strategic decision shifts the security focus to Model Integrity, a key component of AI security that protects the system's logic from tampering. Specifically, robust data validation during the cleaning phase serves as a defense against Data Poisoning, ensuring that no malicious or degraded inputs compromise the model's training or lead to misclassification. Furthermore, the system reinforces security by encrypting all communications and user location data during transmission, aligning with best practices for securing AI lifecycles and maintaining a resilient security posture.

2.3 Model Development Plan

2.3.1 Baseline Model Choice with Justification

The process of detecting vacant and occupied parking spaces involves sophisticated visual recognition and spatial analysis. To achieve high accuracy and efficiency, the proposed system is organized into two primary stages:

Stage 1: Automated Detection and Classification

The first stage focuses on the core computer vision task: identifying vehicles and determining the occupancy status of each parking slot within the facility. In this project, the **YOLO (You Only Look Once)** framework is utilized for the detection process [1]. YOLO is a state-of-the-art deep learning architecture designed for high-performance object detection. It treats detection as a single regression problem, mapping image pixels directly to bounding box coordinates and class probabilities in a single pass.

By employing this neural network approach, the system can capture complex visual features and patterns, ensuring reliable detection even under challenging environmental conditions such as varying lighting, shadows, and different camera perspectives. This automated classification provides a real-time digital map of the parking lot's status, which serves as the fundamental data input for the subsequent recommendation engine.

Stage 2: Nearest Parking Space Determination and Pathfinding

Following the processing of visual data and identification of vacant slots in Stage 1, the system proceeds to the decision making phase. This stage focuses on computing the most efficient route from the driver's entry point to the nearest available parking space.

The system implements an internal representation of the parking layout combined with the *A (A-Star) Search Algorithm**. This algorithm analyzes the parking lot as a grid or network, calculating the shortest traversable path based on the occupancy data generated by the YOLO model. By utilizing A* pathfinding, the system provides a precise and computationally efficient solution for navigating complex parking environments. This approach ensures rapid response times and high recommendation accuracy, guiding the user effectively to the optimal vacant spot.[2]

Stage 2

AI System Design PROJECT Umm Al Qura University



2.3.2 Experimentation Plan with Evaluation Metrics

The proposed system will be evaluated based on two main components:

- Parking space detection and occupancy classification using the YOLO computer vision model.
- Recommendation of the nearest available parking space using the A* pathfinding algorithm.

1. Parking Detection and Classification Evaluation

The ability of the YOLO model to detect and classify parking spaces as available or occupied will be evaluated using standard object detection metrics, including:

- **mAP50:** Measures the model's accuracy at an Intersection over Union (IoU) threshold of 0.5.
- **mAP50–95:** Provides a more comprehensive evaluation across multiple IoU thresholds.
- **Precision:** Indicates how many parking spaces predicted as available are actually available, reducing cases where the system suggests an occupied parking space.
- **Recall:** Measures the ability of the system to correctly detect parking spaces and classify their occupancy status.

2. Nearest Parking Recommendation Evaluation

After identifying the available parking spaces using the **YOLO model**, the system applies the **A* pathfinding algorithm** to generate a route from the user's location to the nearest available parking space. The recommendation module is evaluated by **visually verifying** that the selected parking space is available and that the generated path correctly connects the user's position to the recommended parking spot. The output results are displayed directly on the parking image, including the user location, selected parking space, and **computed route**.

2.3.3 Model Improvement Plan

The performance of the **YOLOv8** model will be improved through an iterative experimentation process by adjusting different training configurations, such as the **number of epochs**, **batch size**, and **confidence threshold**, to achieve better detection and classification accuracy. In addition, **built-in augmentation techniques** in YOLOv8 will be utilized, which automatically modify training images, such as changing **lighting conditions** or **camera angles**, to improve the model's ability to generalize and handle different environmental conditions.

Furthermore, different experimental results will be compared to identify the most accurate and reliable model configuration for parking space **detection** and **occupancy classification**. Additional improvements may include expanding the dataset, increasing the number of training images collected under different weather and lighting conditions, and testing larger YOLOv8 variants to improve the reliability of parking space detection under different environmental conditions and reduce classification errors.

For the nearest parking recommendation module, future improvements may include enhancing the efficiency of the **pathfinding algorithm** and reducing potential errors in selecting the **nearest available parking space**. These improvements aim to make the system more reliable and suitable for real-world smart parking applications.

Stage 2

AI System Design PROJECT Umm Al Qura University



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